User Data and Bootstrapping

Explore how to automate Amazon EC2 instance configuration using user data scripts and bootstrapping techniques. Understand securely integrating instances with AWS services via IAM roles, best practices for scripting, and how this automation supports scalable and maintainable cloud applications.

Automating the setup and configuration process when launching Amazon EC2 instances is a core part of building scalable and maintainable cloud applications. This automation ensures that each instance is consistently configured, regardless of when or where it's launched. Two foundational AWS features that enable this automation are user data and bootstrapping.

AWS developers must understand how to effectively utilize user data and implement bootstrapping strategies to streamline deployments, reduce manual errors, and scale operations reliably. In this lesson, we'll break down the technical mechanisms behind user data, show how bootstrapping works in real-world scenarios, and explain how to securely integrate EC2 instances with other AWS services using IAM roles.

Bootstrapping applications on EC2

Bootstrapping in AWS refers to the automated process of preparing an instance to run specific workloads immediately after it launches. This includes installing required software, configuring services, initializing application logic, and connecting to other AWS services. Bootstrapping reduces setup time, enf consistency, and integrates EC2 instances seamlessly into broader system architectures.

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Unlike prebaked AMIs that contain software already installed, bootstrapped instances pull everything they need at runtime. This makes them highly dynamic and reduces the need to maintain numerous custom AMIs. Additionally, it supports version-controlled infrastructure, where application state, configuration, and deployment logic all live in source-controlled scripts.

User data scripts

User data in Amazon EC2 provides a powerful mechanism for automating the configuration of virtual servers at the time of launch. By injecting scripts or `cloud-init` directives into the instance's lifecycle, we enable dynamic infrastructure behavior that aligns closely with infrastructure as code (IaC) principles.

These scripts are usually written in Bash, PowerShell, or `cloud-init` directives, which EC2 executes during the boot process. This script can install and configure software, pull application code from storage or repositories, register with monitoring agents, or customize the operating system. User data is particularly important in automating infrastructure provisioning, supporting IaC, and enabling integration with other AWS services at launch time.

Here are sample user data scripts for different operating systems.

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```
#!/bin/bash
yum update -y
yum install -y httpd
systemctl enable httpd
systemctl start httpd
```

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User data scripts for Linux

When and how user data is executed

User data scripts are executed automatically during the first boot of the EC2 instance. By default, this is a one-time operation, which helps ensure consistent and predictable initialization. However, certain use cases may require the script to run on every reboot or relaunch. For these scenarios, we can configure the instance with `cloud-init` directives to reprocess user data across boots. This makes it possible to maintain consistent behavior, especially when launching fleets of instances or recovering from instance stops and starts.

- The `cloud-config` directive allows us to define reusable logic, such as installing packages or starting services.
- The `cloud-init` directive configure the EC2 to always run by modifying the configuration.

This ensures the instance behavior is not just defined at launch but also repeatable across its lifecycle.

How to provide user data at launch

User data is most commonly specified during the EC2 launch process. In the AWS Management Console, we can add user data under the “Advanced details” section of the launch wizard. When using the AWS CLI, we pass user data using the `--user-data` flag.

> **Note:** When using the CLI, the `--user-data` script must be a base64-encoded string unless we use `file://` to read a file directly.

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Additionally, user data can be embedded into infrastructure templates using services like AWS CloudFormation, AWS CDK, or Terraform, enabling full automation across provisioning pipelines.

Here's an example of using the AWS CLI to launch an Amazon Linux instance that installs Python and dependencies from an S3-hosted requirements file:

cli-command.sh Shell-20

```shell
aws ec2 run-instances \
  --image-id <valid_ami_ID> \  # Replace with appropriate AMI ID
  --instance-type t2.micro \
  --key-name MyKeyPair \
  --security-groups MySecurityGroup \
  --user-data file://main.sh \
  --iam-instance-profile Name="EC2RoleWithS3Access" \
  --region us-east-1
```

User data script

The user data script is OS-dependent, as the above script works with Amazon Linux. The above user data script in `main.sh` updates the system packages, installs Python, and the requirements from `requirements.txt` file stored in an S3 bucket, `mybucket`.

> **Note:** To access the `requirements.txt` file, or interact with any AWS service, the EC2 instance must be assigned an IAM role with the appropriate permissions. This ensures the instance can perform the required action such as reading objects from S3.

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Common use cases for bootstrapping

Bootstrapping is essential in a variety of real-world cloud deployment scenarios. For instance, web servers are often bootstrapped to install and start services like Apache, NGINX, or IIS. Applications that rely on backend logic may need to download source code from Git repositories or Amazon S3 during the boot phase. Similarly, EC2 instances can be configured to join a load balancer, register with an Auto Scaling group, or start background services like database agents or logging daemons.

Here are some of the most common tasks automated through bootstrapping:

- Installing web or application servers (e.g., Apache, Node.js, and IIS)
- Downloading and deploying application code from GitHub, CodeCommit, or Amazon S3
- Installing system dependencies or runtime environments
- Applying OS-level configurations such as firewall rules or sysctl parameters
- Registering with monitoring services like Amazon CloudWatch agent or third-party log aggregators
- Connecting to configuration management tools (e.g., AWS Systems Manager, Ansible)

main.sh Shell

```
#!/bin/bash
sudo yum -y update &&\
sudo yum -y install git &&\
curl -fsSL https://rpm.nodesource.com/setup_21.x | sudo bash -
sudo yum install -y nodejs && sudo dnf install nodejs -y &&\
git clone https://github.com/Educative-Content/aws-compute-services-z
cd aws-compute-services-zero-to-hero-cl-frontend-app &&\
npm install &&\
npm start
```

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Bootstrapping EC2

The above user data script works with Amazon Linux, and updates the system packages, installs Git and Node.js, clones a Git repository containing a Node.js application, installs the application's dependencies, and starts the application.

> **Note:** When launching multiple instances with identical configurations, using a custom AMI or a launch script ensures consistent deployments. Relying on manual user data configuration can lead to errors due to differences in base AMIs or other environment variations.

Benefits of bootstrapping over preconfigured AMIs

Bootstrapping provides several advantages compared to relying solely on custom-built AMIs:

- **Flexibility:** Different environments (e.g., dev, staging, and prod) can be configured from the same base image with minimal changes to the script.
- **Speed of updates:** New software versions or configuration changes can be applied instantly without rebuilding and redistributing an AMI.
- **Simplified maintenance:** There are fewer AMIs to manage and test across the fleet.
- **Consistency:** All instances derive their behavior from version-controlled scripts, ensuring predictable environments.

However, for applications with heavy dependencies or strict performance requirements at boot, using a hybrid approach, bootstrapping a lightweight configuration while leveraging a prebuilt AMI, can offer a balance between and flexibility.

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Why avoid hardcoded credentials?

Hardcoding AWS access keys into EC2 instances presents significant security risks. These credentials may be inadvertently exposed through logs, source code repositories, or AMI snapshots. Worse, if leaked, they can be exploited by unauthorized actors to access our AWS resources. Managing and rotating these keys also becomes a manual, error-prone process.

In contrast, IAM roles for EC2 offer a secure, built-in mechanism to access AWS services without storing credentials directly. When an IAM role is attached to an EC2 instance, AWS automatically delivers temporary, short-lived credentials to the instance via the Instance Metadata Service (IMDS). These credentials are rotated automatically and scoped by policies, reducing the blast radius of potential misuse.

Accessing AWS resources using instance profile

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How IAM roles work on EC2

IAM roles are designed to be assumed by AWS services. When associated with EC2 instances, the role is assumed automatically at launch. The temporary credentials associated with that role are then accessible inside the instance via IMDS, which now defaults to the more secure IMDSv2. IMDSv2 requires session-oriented requests and mitigates common attacks such as server-side request forgery (SSRF), offering a more hardened security posture compared to IMDSv1.

To use an IAM role from inside the instance, applications retrieve credentials from the metadata endpoint (or automatically if using an AWS SDK like Boto3), and the SDK transparently injects these into service calls.

Here's how to launch an EC2 instance with an IAM role using the AWS CLI:

main.sh Shell

```shell
aws ec2 run-instances \
  --image-id <Valid_AMI_ID> \
  --instance-type t2.micro \
  --key-name MyKeyPair \
  --iam-instance-profile Name="MyEC2Role" \
  --user-data file://install-s3.sh \
  --region us-east-1
```

Launch EC2 with IAM Role

In this example, the IAM role `MyEC2Role` must already exist, and it must include permissions for the actions the instance will perform, such as `s3:GetObje`

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> **Note:** IAM roles must be attached at the time of EC2 instance creation, or later when the instance is in a stopped state. This is a critical detail, roles

> can be attached or changed while the instance is running. However, applications using the instance metadata service may continue using cached credentials, so the role change may not take effect in the application until credentials are refreshed or the application is restarted.

Once the IAM role is attached, we can safely call AWS services from inside the user data script using an SDK like Boto3 for Python. The credentials are injected automatically from the instance metadata, so no explicit authentication setup is required.

main.sh Shell

```shell
#!/bin/bash
yum update -y
yum install -y python3 pip
pip3 install boto3

cat <<EOF > /home/ec2-user/fetch_s3.py
import boto3

s3 = boto3.client('s3')
response = s3.get_object(Bucket='my-app-data', Key='config.json')

with open('/home/ec2-user/config.json', 'wb') as f:
    f.write(response['Body'].read())
```

Accessing S3 from a Boot script

This script, referenced while creating the instance, installs Python and Boto3, creates a Python script that fetches a file from an S3 bucket using temporary credentials provided by the IAM role, and saves it locally. The instance automatically has access to S3 through the role, no credentials are hardcod exposed.

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Common pitfalls and debugging tips

Even though user data and IAM roles provide streamlined automation and security, issues can occur, especially during first implementations or migrations. Understanding the typical failure points and knowing where to look can save hours of debugging and help avoid unnecessary instance relaunches.

Why isn't user data executing?

One of the most frequent problems developers face is a user data script that seemingly doesn't run. Fortunately, this is usually easy to diagnose by checking the right logs.

For Linux instances, the primary logs for user data execution are as follows:

- `/var/log/cloud-init-output.log`**:** This captures the actual output (both successes and failures) from the script.
- `/var/log/cloud-init.log`**:** This provides `cloud-init` process details and errors, including parsing issues or syntax errors in `cloud-config`.

For Windows instances, user data execution logs are located at: `C:\ProgramData\Amazon\EC2-Windows\Launch\Log\UserDataExecution.log`

If a script doesn't execute, double-check these common mistakes:

- The script must begin with a proper shebang (`#!`) line. For example, `#!/bin/bash` for Bash, `#!/usr/bin/env python3` for Python, or `<powershell>` for Windows.
- Permission issues can prevent services from starting. Ensure correct paths, users, and service names are configured.
- Silent failures may result from improper syntax or inaccessible resources. Use `set -x` in Bash to trace execution.

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Why isn't the IAM role working?

IAM roles are powerful, but they must be properly configured and attached correctly. If our instance cannot access AWS resources (like S3 or DynamoDB), we can start by verifying the following:

- **Role attachment:** Confirm the EC2 instance has a role attached. We can inspect this from the console (under “Instance Settings”) or by running the following command:

main.sh Shell

```
curl http://169.254.169.254/latest/meta-data/iam/info
```

> **Note:** The IP address `169.254.169.254` is the EC2 instance's gateway to learning about itself and, most crucially for secure development, to obtaining the IAM permissions it needs to interact with other AWS services without embedding long-term credentials.

- **IAM policy permissions:** The role’s permissions must explicitly allow the action on the resource. Use the IAM policy simulator or review JSON policies to confirm.
- **Trust policy configuration:** Ensure the trust relationship includes:

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index.json Node.js

```
"Principal": {
  "Service": "ec2.amazonaws.com"
}
```

Principal policy for roles associated with EC2

Without this, EC2 cannot assume the role.

- **IMDS access:** Ensure that the instance can access the Instance Metadata Service. If using IMDSv2 (recommended), scripts must use session-oriented requests, which may require tool updates.

Bonus pro tips

Here are some extra tips on manually re-invoking `cloud-init` to rerun user data post-launch:

- `cloud-init` **reruns:** If we need to rerun user data after launch, `cloud-init` must be manually re-triggered. The commands below can be used within the EC2 instance to rerun the launch script:

main.sh Shell

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```
sudo cloud-init clean
sudo cloud-init init
sudo cloud-init modules --mode=config
sudo cloud-init modules --mode=final
```

Execute the launch script from the instance

- **Testing safely:** Always test scripts in a non-production environment. Consider using EC2 spot instances for quick, cost-effective testing.
- **Modular scripts:** Break long user data scripts into separate, well-commented sections or source external files using `curl` or `aws s3 cp` for maintainability.

By understanding these pitfalls and implementing robust validation during the launch process, we can reduce deployment failures, shorten troubleshooting time, and increase confidence in our EC2 automation workflows.

Conclusion

In this lesson, we explored how to automate and securely configure Amazon EC2 instances using user data scripts and IAM roles, key tools for infrastructure as code workflows. We covered how user data enables instance bootstrapping, common use cases like software installation and pulling code from S3, and how IAM roles allow secure access to AWS services without hardcoding credentials. We also discussed best practices, debugging tips, and common pitfalls to ensure reliable automation.

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